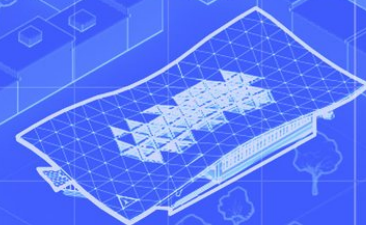
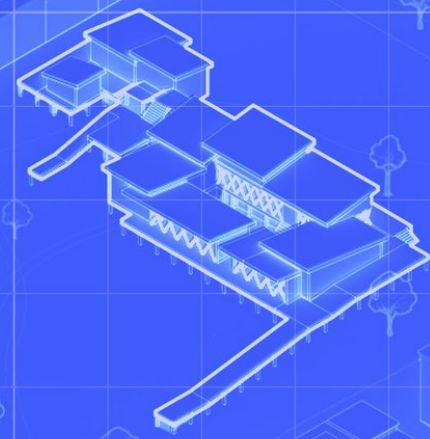
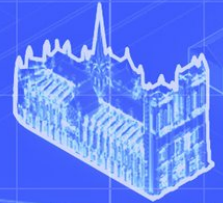
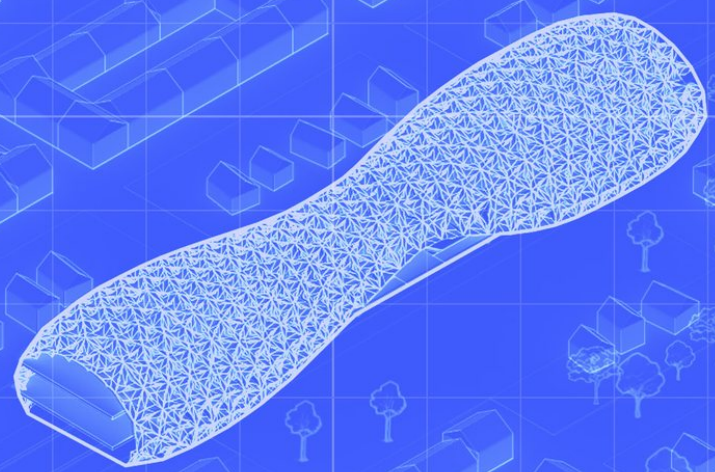


Portfolio Simon Hildell

Selected works 2026





Who I am

My name is Simon Hildell. I'm 24, originally from Gothenburg, Sweden. I'm currently working at Finch in Malmö, on a break from an M.Sc in digital architecture I'm pursuing in Lund I'm a curious guy who likes understanding the actual logic behind things, which is probably why I've always had a strong interest in mathematics, and why that eventually pulled me toward computational design. I enjoy communicating my ideas through whatever medium fits best, complex digital models, quick sketches, physical models and prototypes, sometimes all three on the same project.

Outside of architecture I'm into movies, boardgames and quiz nights. I am a big brother of three girls and have played football for most of my life. In this portfolio I've gathered some of my favourite projects, the ones I most want to share with you.

What I do

I like running into problems, my own and other people's. I'm curious about how things actually work, I want to get to the bottom of the logic behind whatever I'm looking at, not just get it to function. A lot of that curiosity turns into automating things, because once I understand the logic well enough to automate it, that frees up time, mine or someone else's, for the parts of the work that are actually fun or important.

Solving the problem and building the automation for it are both things I genuinely enjoy. I stay super up to date with the newest tools available and beyond my in-depth knowledge in most of the main tools used in architecture, I have worked with applied AI, know how to work with mcp servers and most other tools in the realm of Architecture, engineering, construction and software.

Experience

Architect / Computational Designer
Finch 3D, Jun 2026 - Dec 2026
Internship

Polestar specialist
Polestar, Dec 2022 - Jul 2025
Extra and summer work

Platoon leader
Swedish armed forces, Jun 2021 - Aug 2022
Military service

Education

Master of Science in Digital Architecture and Emergent Futures
Lund University, LTH, Aug 2025 -
60/120hp completed (Study break)

Bachelor of Fine Arts in Architecture
Umeå School of Architecture, Aug 2022 - June 2025
180hp

Additive Manufacturing
Chalmers University of Technology, Summer 2025
5hp, FDM, PBF, SLA, MJT, BJT

Mathematical Modelling and Data Processing in Python
Malmö University, Summer 2025
7.5hp, data processing and visualization

Cloud-Based 3D CAD
University of Borås, Summer 2025
7.5 hp, CAD design and FEA simulations

Construction CAD Continuation course
Dalarna University, Spring 2025
7.5 hp, complex Revit modelling and documentation

Construction CAD
Dalarna University, Fall 2024
7.5 hp, Revit and building permit drawings

Calculus
University of Gävle, Fall 2024
7.5 hp, Single variable calculus

Linear Algebra
University of Gävle, Spring 2024
7.5 hp, vectors, matrices and transformations

Algebra and Geometry
University of Gävle, Fall 2026
7.5 hp, mathematical structures and vectors

Natural Science Programme
Hulebäcksgymnasiet, 2018 - 2021
Final grade 21.25/22.5

Software

Advanced
Rhino
Grasshopper
Revit

Competent
Illustrator
Photoshop
Rhino.Inside.Revit

Intermediate
InDesign
3DS Max
Corona
Python
After Effects

Worth mentioning
rhinomcp
Revit MCP
API integration
Claude and Codex integrations

Awards

URBFORM 1:1 Competition winner
Won a student competition with entry "Campustratten",
included in portfolio, to be built.

Sports Soldier of the year
Diploma from Swedish Armed Forces for honourable
achievements connected to sports.

Languages

Fluent
Swedish
English

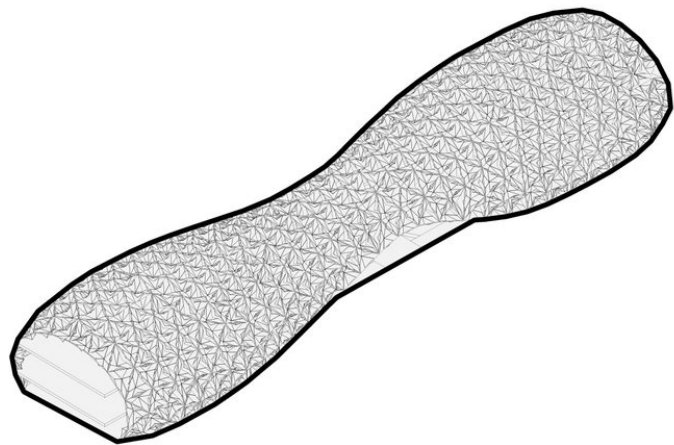
Conversational
Spanish
Danish

Contact

Direct
Email: simon@hildell.com
Phone: +46 708 96 36 05
LinkedIn: www.linkedin.com/in/simon-hildell-19b078107

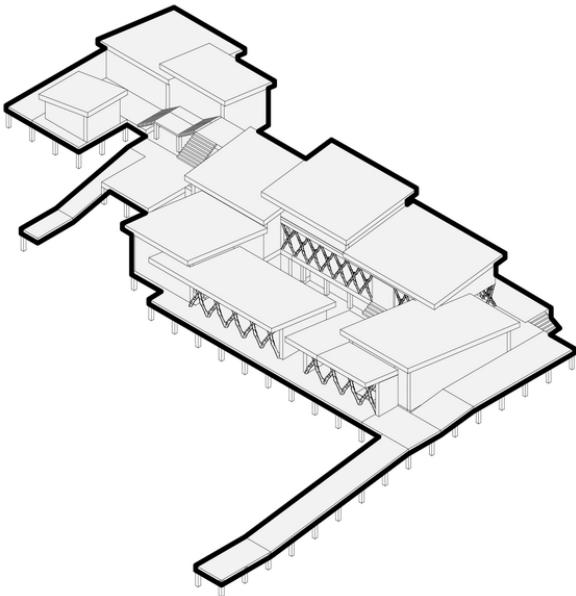
More
Website: www.simonhildell.com
Instagram: [@simonhildellworks](https://www.instagram.com/simonhildellworks)

Zephyr



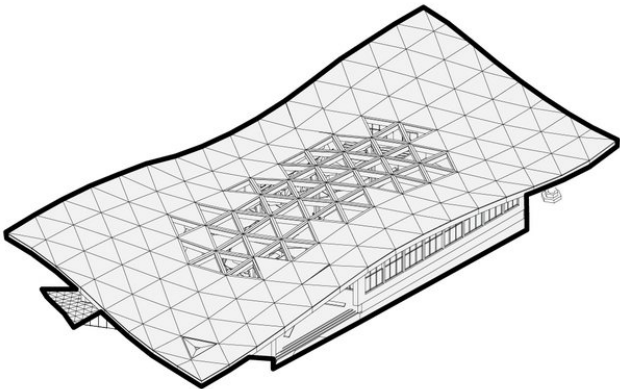
Spring 2026
Academic group project, LTH
Pages 03-05

Naturum



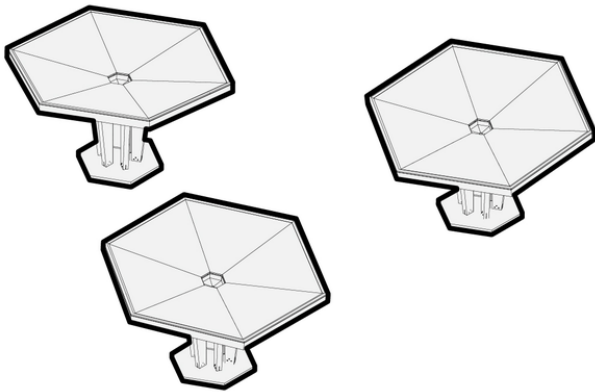
Fall 2025
Academic duo project, LTH
Pages 06-08

Rain hub

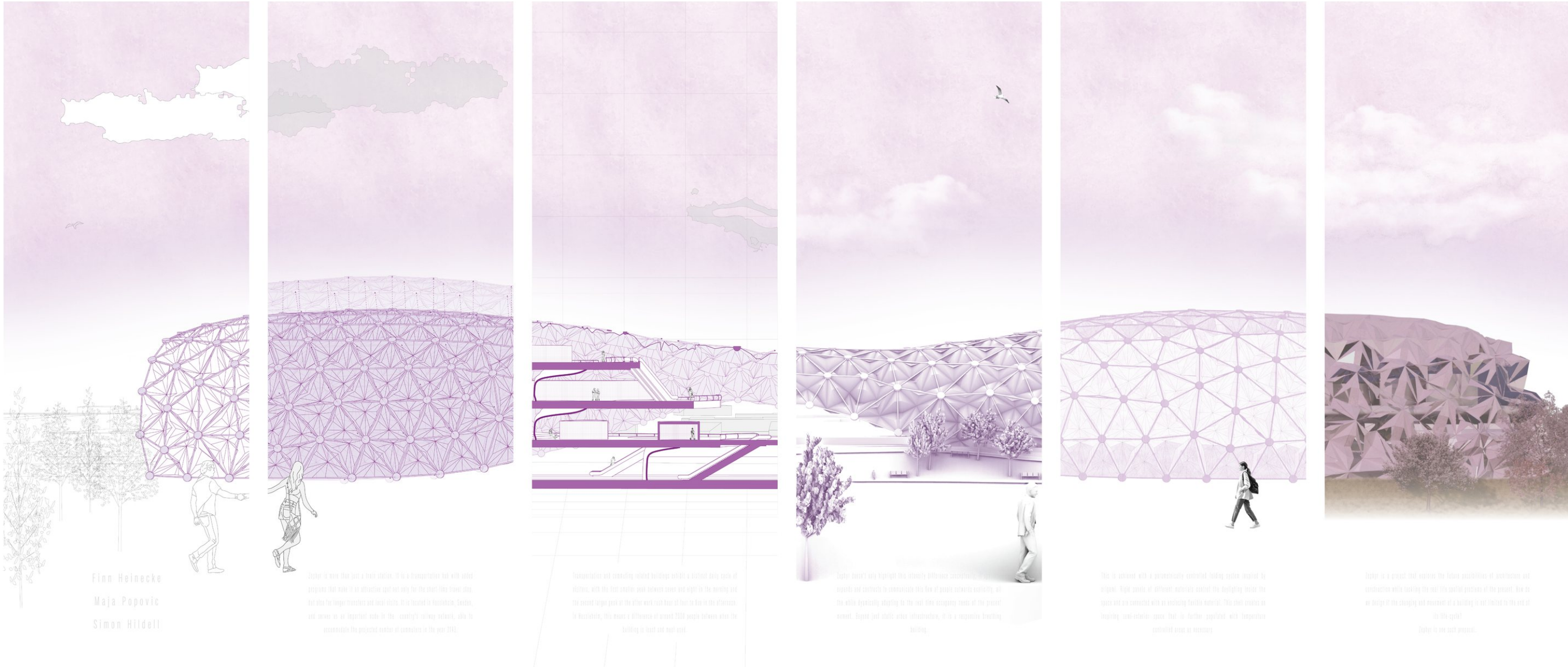


Spring 2025
Individual bachelor project, UMA
Pages 09-12

Campustratten



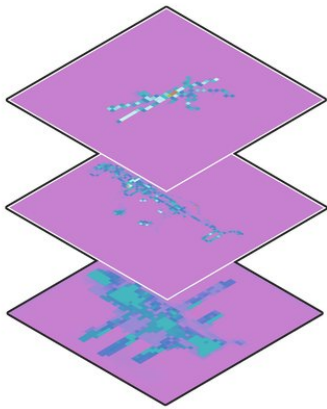
Summer 2026
URBFORM 1:1 Competition winner - duo entry to be built
Pages 13-15



Zephyr

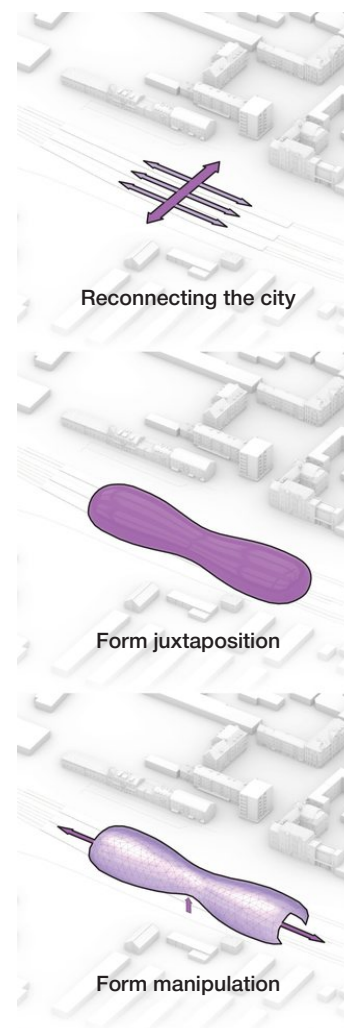
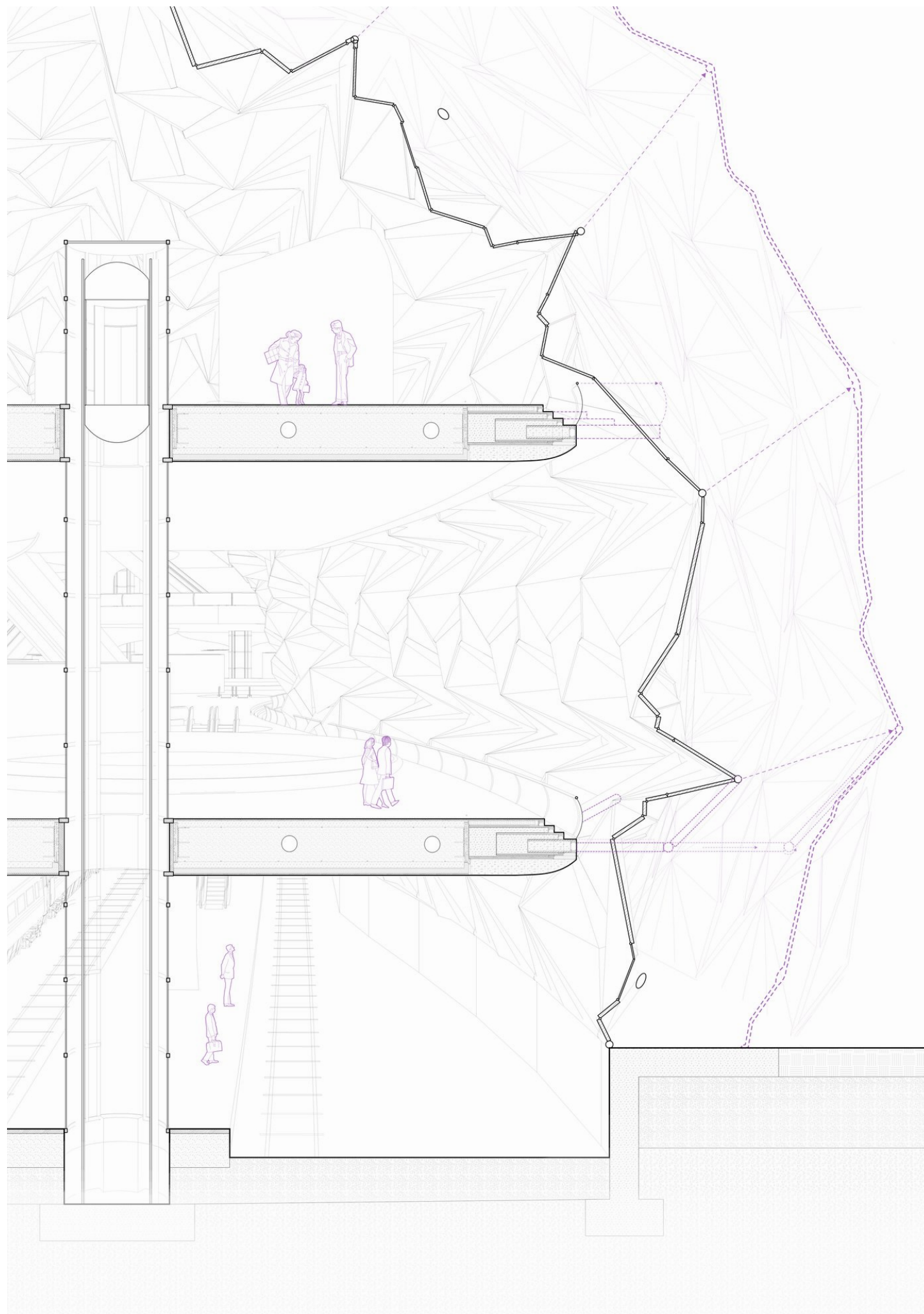
Spring 2026
Academic group project, LTH

A breathing transport hub in Hässleholm. We developed a panel system rooted in origami which is able to geometrically breathe, expanding and contracting in three dimensions as a response to the people moving through it. Based on projections for human traffic in the hub over the coming years, we mapped passenger flow across a full week and let that data sculpt the structure across time. We started from an effectivity standpoint, trying to optimize the space, but ended up somewhere else entirely. The geometry stopped solving a flow problem and started just showing it. Not to speed anyone up or make the station work better, but to acknowledge the people inside it. It's a quiet gesture, but every expansion in the panel system traces back to a real, measured moment of people moving through the space.



Optimization as a startpoint

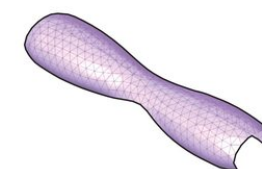
The breathing came from data, not a feeling. We simulated empty stations with people walking through them, and as they moved each one left a point behind it, a trail tracing exactly where they'd been. Over a full simulation those trails built up into heatmaps, and we used those heatmaps as a ground to draw curves on. Based on the density we decided how large the shape in the panel system should grow at that point, though calling them blobs is maybe too clean. They were closer to non defined shapes, loose forms that just followed wherever the trails pooled up. Busier zones grew more, quieter corners stayed close to neutral. It started from an effectivity standpoint, we wanted the geometry to actually respond to congestion. But the more we worked with the trails the less it was about solving anything. It turned into an art piece, a structure whose only real job is to show how many people were there. Nothing in the final geometry is random even though the result reads as sculptural rather than functional, and that's kind of the point.



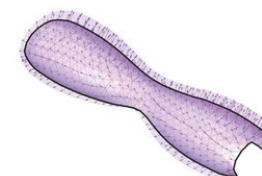
Reconnecting the city

Form juxtaposition

Form manipulation



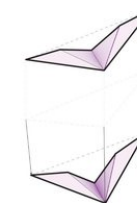
Minimal mesh



Growth direction



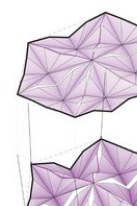
Breathing intensity



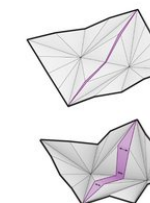
Rigid folding principle



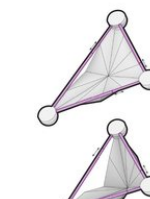
Unit cell



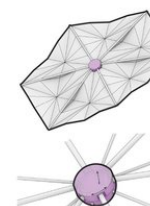
Cluster



Enclosing with flexible material



Expanding pipe frame

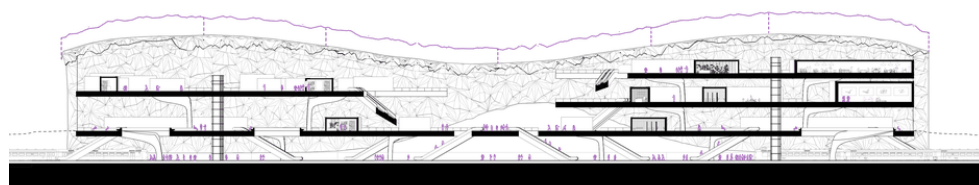


Moving joint

Breathing system

The whole thing started with origami. We wanted to understand how a folded surface can expand and contract without tearing or locking up, so before touching Grasshopper we tested a bunch of different origami software just to get a feel for the logic, both physically and digitally. We folded paper and card by hand alongside that to see what actually worked in real material, not just on screen.

Once we understood the fold logic we moved it into Grasshopper, where we could scale the same principle up into shapes way more complex than we could fold by hand. The order mattered to us. We didn't design a building first and then look for a system to wrap around it, we wanted to start with the system. The panel logic came first, tested in both physical models and different software, and the shape of the building grew out of what that system could actually do. Our main principle ended up being three parts working together: solid rigid panels doing the actual folding, a structural pipe system holding the whole thing up, and a flexible material filling the gaps between panels so the surface could move without anything tearing.



Acknowledging the public

What started as an effectivity project ended up as an art piece, and this is where that becomes most visible. The building is really two shapes breathing between each other, and from a distance that breathing becomes a kind of readout. See it expanded and you know the station is busy. See it pulled in close and you know it's quiet. You don't need to be inside or look at a screen, the shape itself tells you. Stay long enough and you'd see it change in real time, contracted in the early morning, swelling out through rush hour, settling back down as the day empties out. It's not showing a number, just a feeling of how full the place is at that moment. We mapped that traffic across a full week to understand where people actually moved and when, and let it carry through into how the two shapes breathe against each other. Not to optimize the space. Just as a gesture. The building is quietly acknowledging the people inside it.





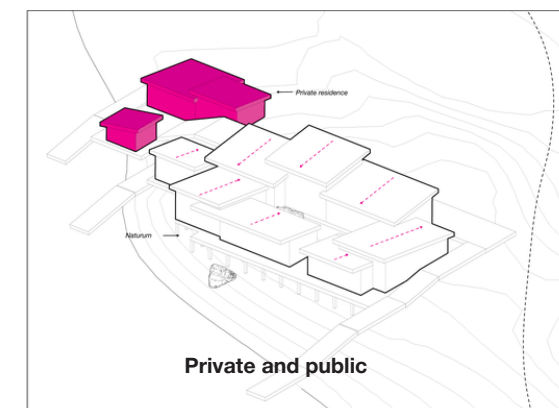
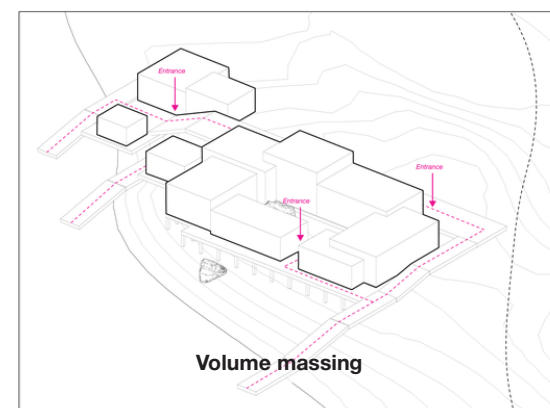
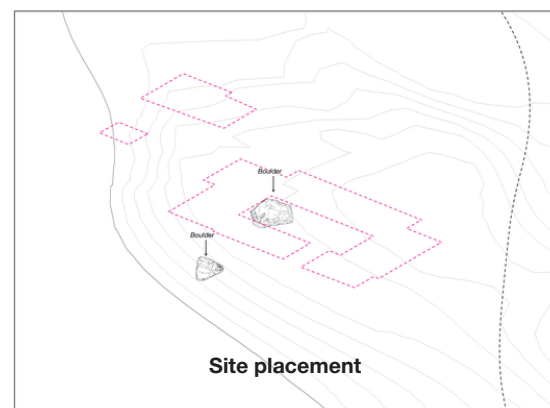
Naturum

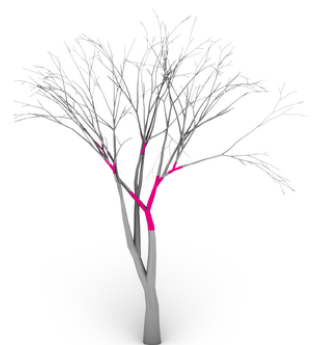
Fall 2025

Academic duo project, LTH

The task was to design a hyperlocal naturum in Breanäs, in north-eastern Skåne. It meant building with what the site itself was already producing and discarding, not just using local materials in the usual sense. Breanäs is an area currently being studied for more sustainable forestry, so the setting made us think carefully about resource use, biodiversity and long-term impact instead of relying on standardized construction systems.

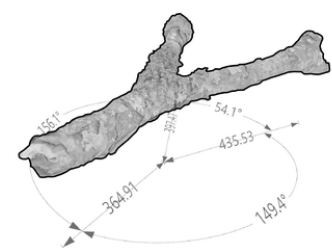
We looked into experimental wood building methods as a starting point, and two references stuck with us the most, *The Forest Datum* by the AA and *Un_Log Factory* by IAAC. Our main objective became designing with a larger part of the tree than usually gets used, building with the Y forks and bent and crooked branches that the forest industry normally throws away. This exploration drove the whole process, both how we figured out how things worked structurally and how we actually ended up designing and shaping the building.





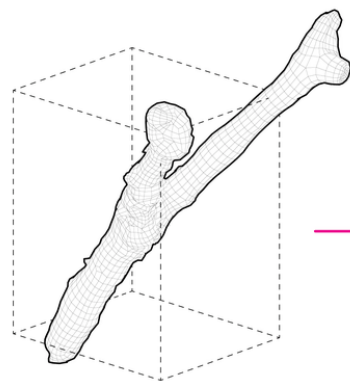
Process

Our main objective was to design with a larger part of the tree compared to what is normally used by designing with what is normally waste. We split the exploration into two parts, the Y fork truss system and straightening logs.



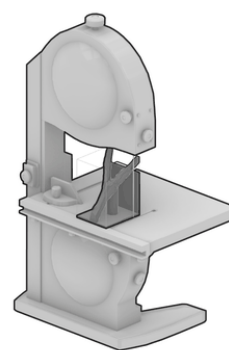
Analyze

The forks are analyzed and based on its geometry a simplification of the Y shape is made. The Y shape is stored in a library.



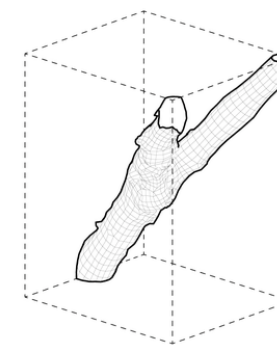
Match up

To fit in the system, the Y forks need to match with the other forks in their row. From the library forks that match in size are matched up.



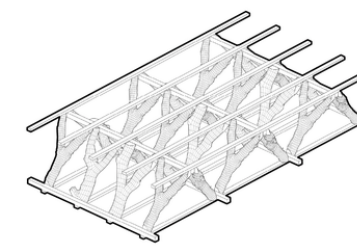
Cut

The Y forks are placed in a jig and placed in a band saw. The method makes the workflow accessible and straightforward.



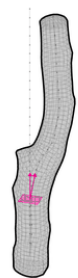
Align

After each fork is cut they are placed back with its matching friends and ready for assembly within a straight grid.



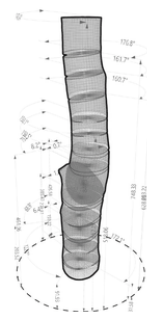
Assemble

The forks are placed in a grid of beams and assembled into a three-dimensional truss system.



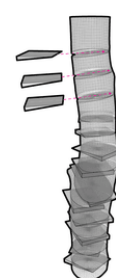
Analyze

The centerline of bent and crooked logs are compared to a straight line and if the tangent deviates more than a threshold, a wedge is inserted.



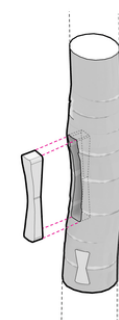
Slit placement

The positions for the slits are described by their height and rotation degree from a reference point to transfer digital information to reality.



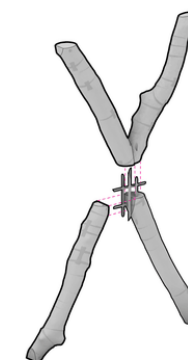
Insert wedges

Wedges are inserted in the slits which alters the geometry of the log and effectively straightens the logs.



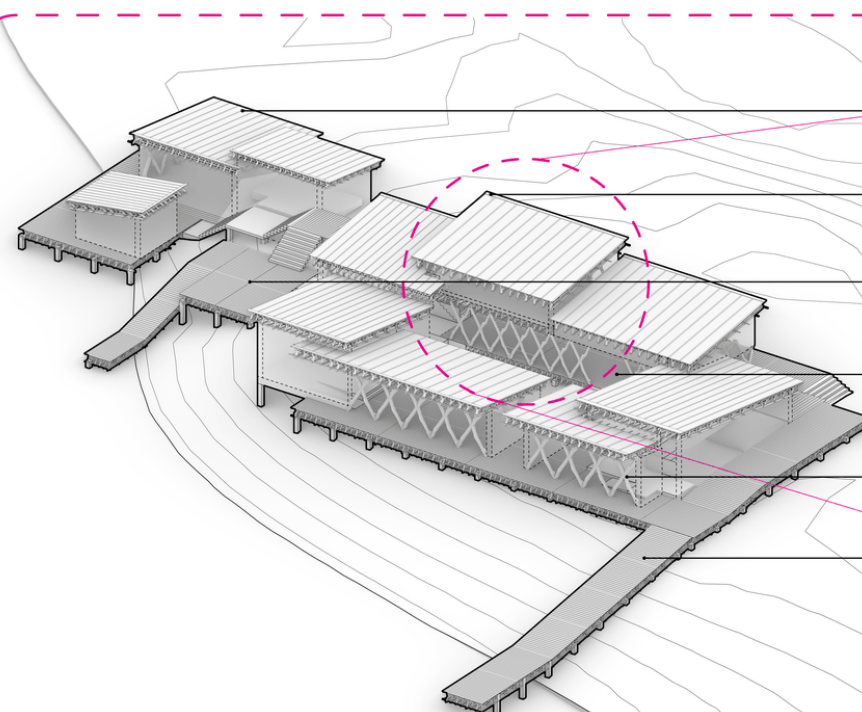
Butterflies

To make up for some of the strength the long fibers gave the log, butterfly keys are inserted over the "open" side of the wedge joint.



Assemble

After the logs are straightened they are assembled in a cross system with dowels as the connections in the middle.



Private residence

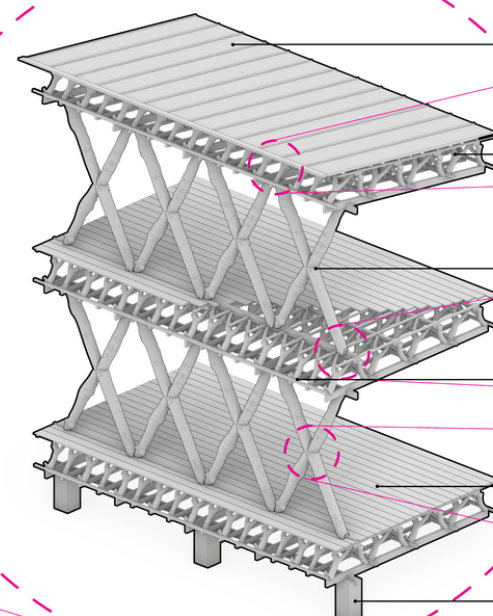
Classroom

Workshop

Cafeteria

Exhibition

Jetty



Metal panel roof

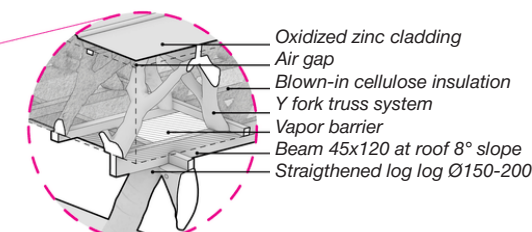
Slab insulation

Cross truss system

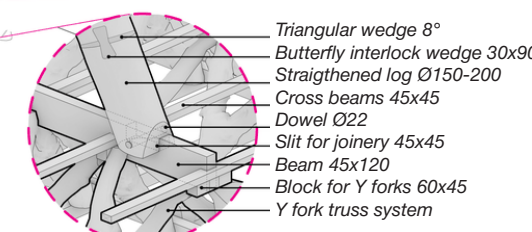
Fork truss slab

Floor

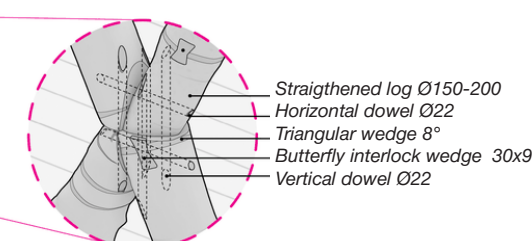
Ground post



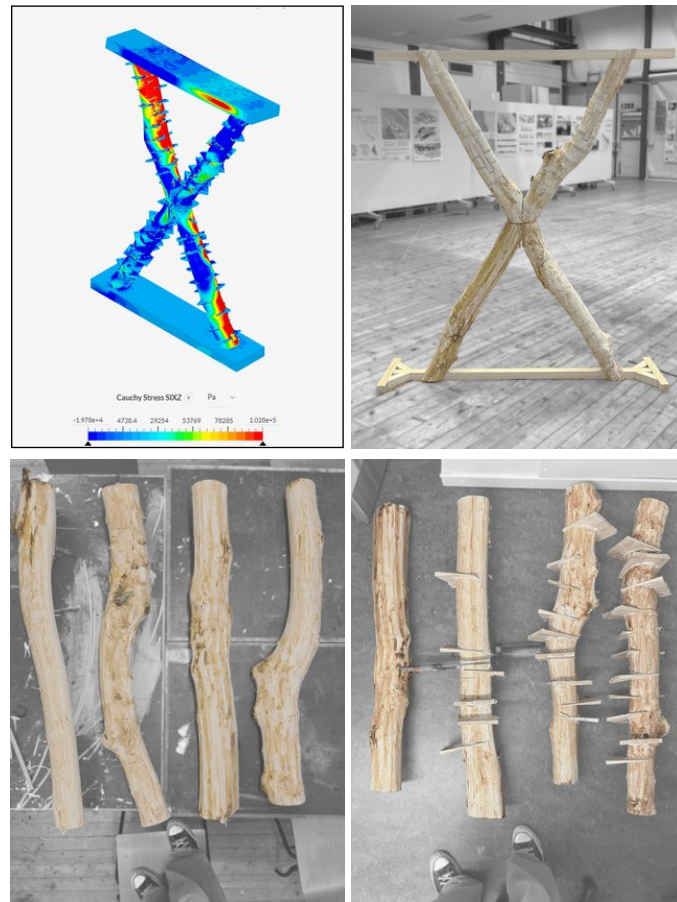
Oxidized zinc cladding
Air gap
Blown-in cellulose insulation
Y fork truss system
Vapor barrier
Beam 45x120 at roof 8° slope
Straightened log log Ø150-200



Triangular wedge 8°
Butterfly interlock wedge 30x90
Straightened log Ø150-200
Cross beams 45x45
Dowel Ø22
Slit for joinery 45x45
Beam 45x120
Block for Y forks 60x45
Y fork truss system



Straightened log Ø150-200
Horizontal dowel Ø22
Triangular wedge 8°
Butterfly interlock wedge 30x90
Vertical dowel Ø22



Prototyping

Getting the straightening method to actually work meant constantly moving between the physical and the digital, neither one was enough on its own. We'd collect bent and crooked logs, cut them into workable pieces and scan them, then feed the mesh into a Grasshopper script that told us exactly where to cut. We cut according to the script and inserted wooden wedges to straighten the log out, and that's where we learned the most, how far a log could actually bend before it split, and how deep we could cut without losing structural integrity. None of that was knowable from the screen alone. We had a working digital script before we ever built anything physical, but the input variables were guesses. Once we had real measurements from cutting and straightening actual logs, we fed them back into the script and it got sharper, closer to what the material was really doing. The two sides kept correcting each other the whole way through, the prototype taught the script and the script guided the next prototype.



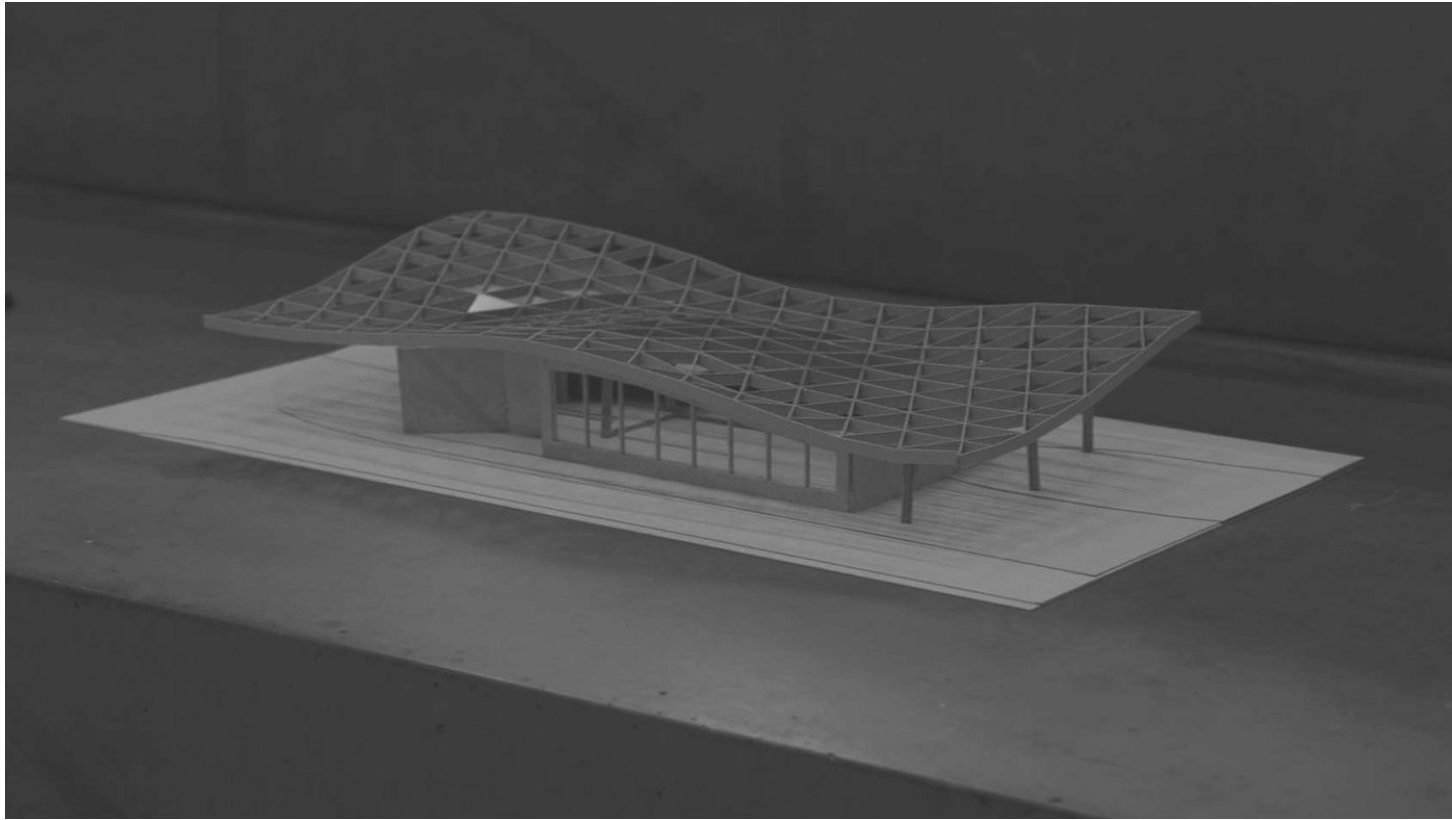


Rain hub

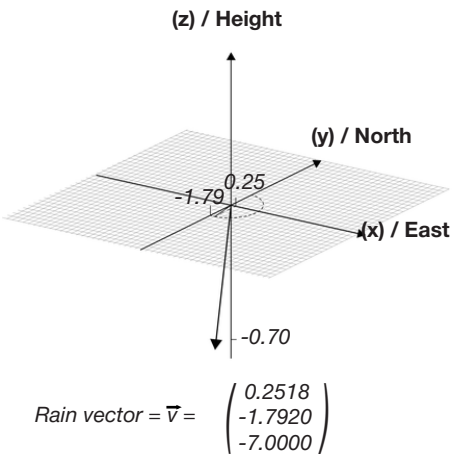
Spring 2025
Individual bachelor project, UMA

Historically in Gothenburg it has rained every third day, and that's not getting better, it's just getting less predictable. The standard response to rain is to avoid it, activities get cancelled or pushed indoors, but not everyone has equal access to a dry alternative.

This project focuses on teenagers specifically, a group with limited access to free, rainproof public space to begin with. Rain Hub is a public building for the activities that normally disappear the moment it starts raining. Instead of shutting the rain out, the building lets it in. Rain becomes part of the experience itself, feeding into elements that add sound, movement, visibility and touch rather than just getting drained away. It grew out of site visits around Gothenburg, conversations with youth leaders at existing youth centers, looking at how teenagers actually use urban space, and research into materials and structures that respond to rain instead of resisting it.

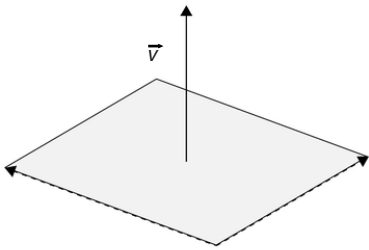


Average rain direction



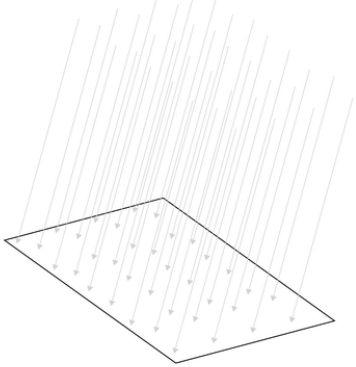
Weather station Järnbrottsmotet
Average vind speed: **1.8069 m/s**
Average vind direction: **172° (360° = N)**
Average rain downward velocity: **-7 m/s**
Data source: **SMHI**

Plane normal

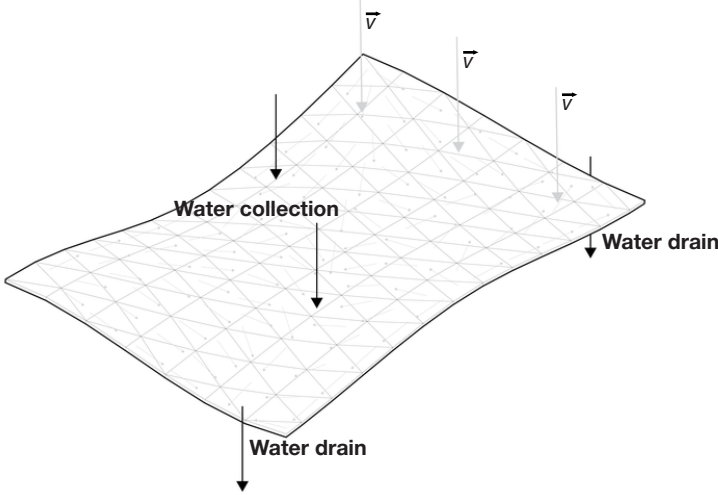


Rain vector orthogonal to surface
Rain hits surface at 90 degrees

For the rain

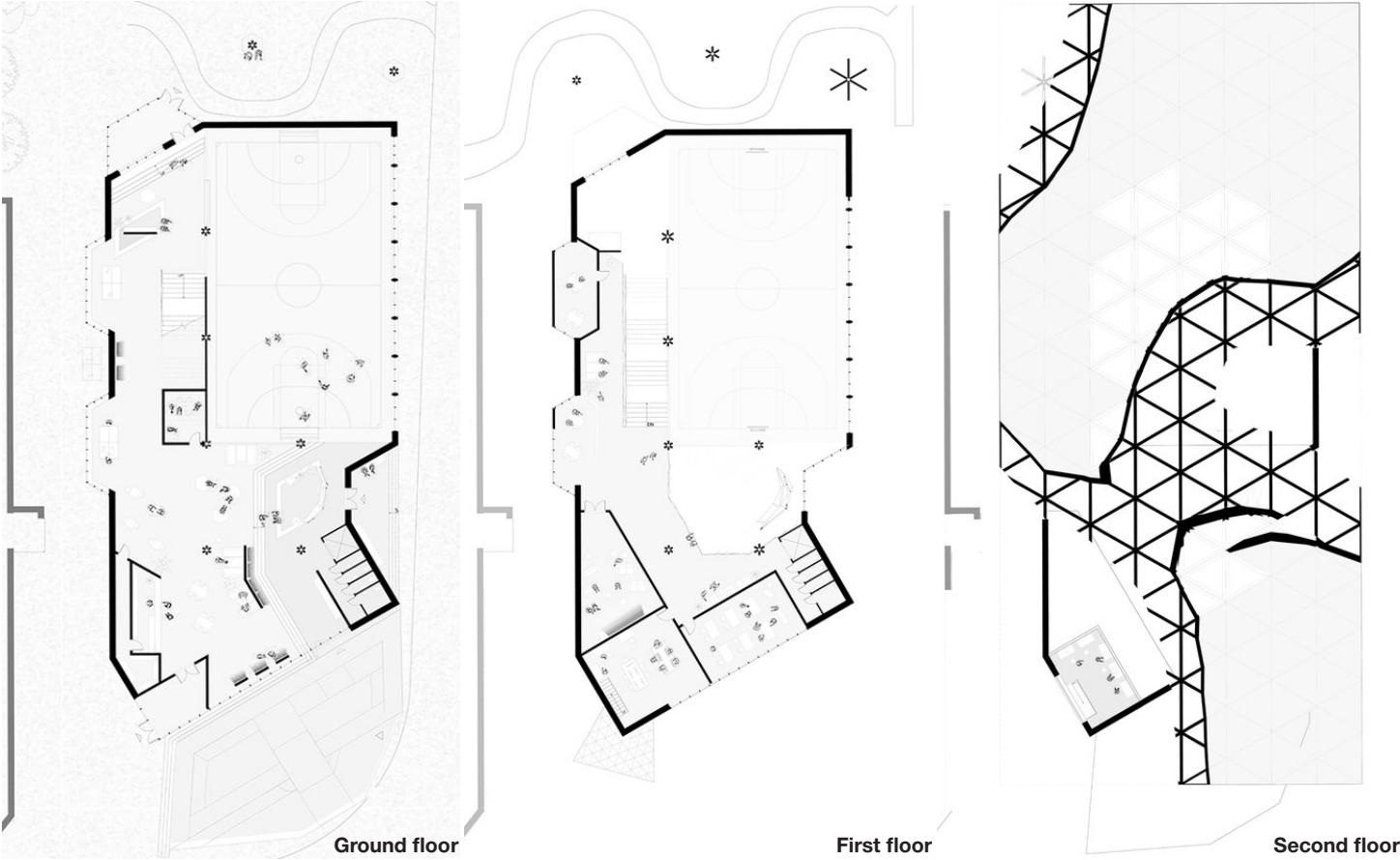


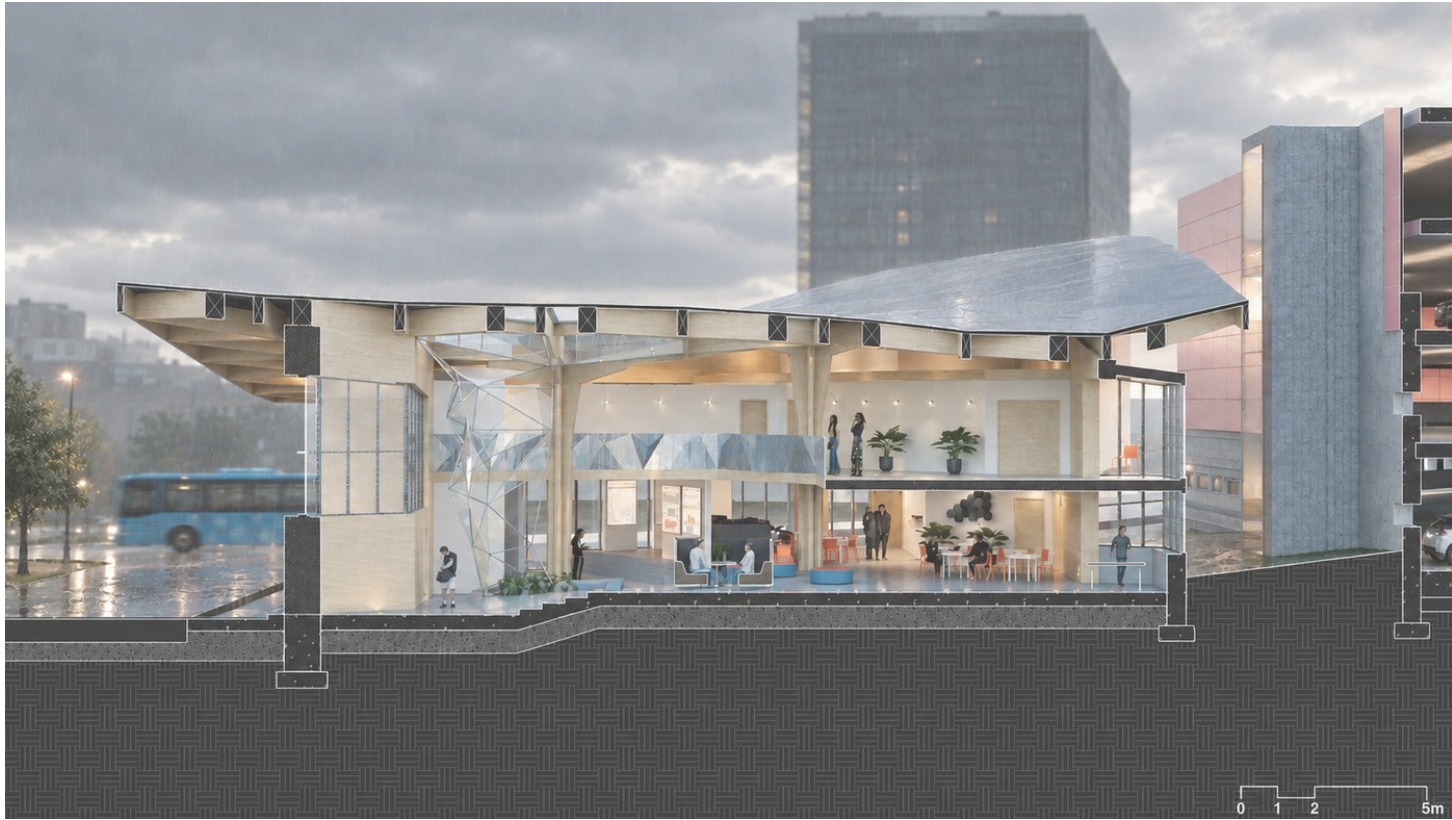
Site specific surface for rain



Rain as an asset

I analyzed the direction rain falls in on site to get an actual rain vector, then used it to orient different parts of the roof depending on what I wanted from them. Some sections are angled to collect as much water as possible, others are angled so the rain hits them orthogonally, which makes the sound as loud as it can get. That sound became its own kind of space, loud enough underneath to feel private even in the open. The roof itself is organically shaped, its form coming directly out of that rain analysis, but structurally it runs on a triangular grid, and that grid ended up shaping everything underneath it, aligning the structure and the programme to whatever the roof was doing at that point.

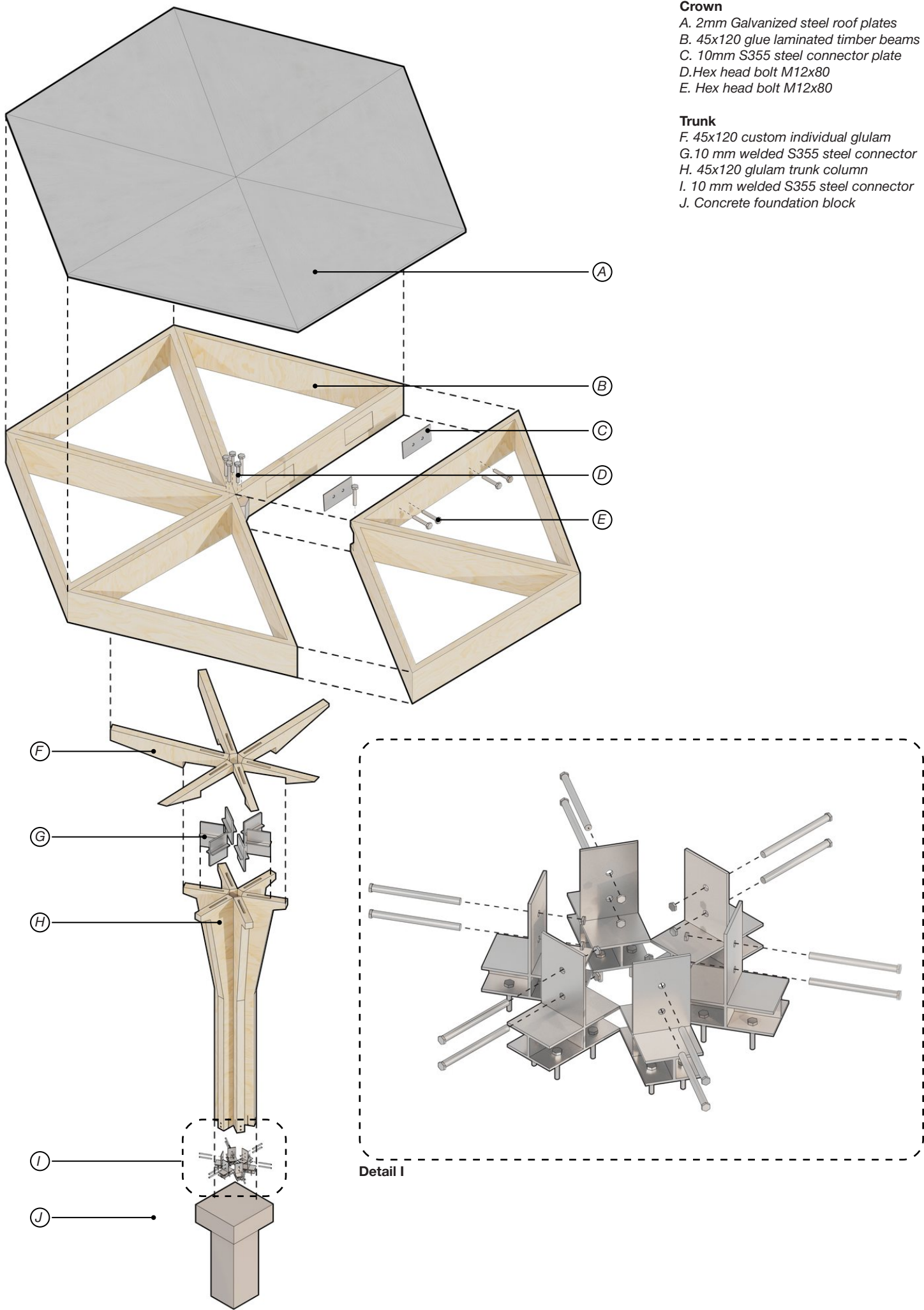


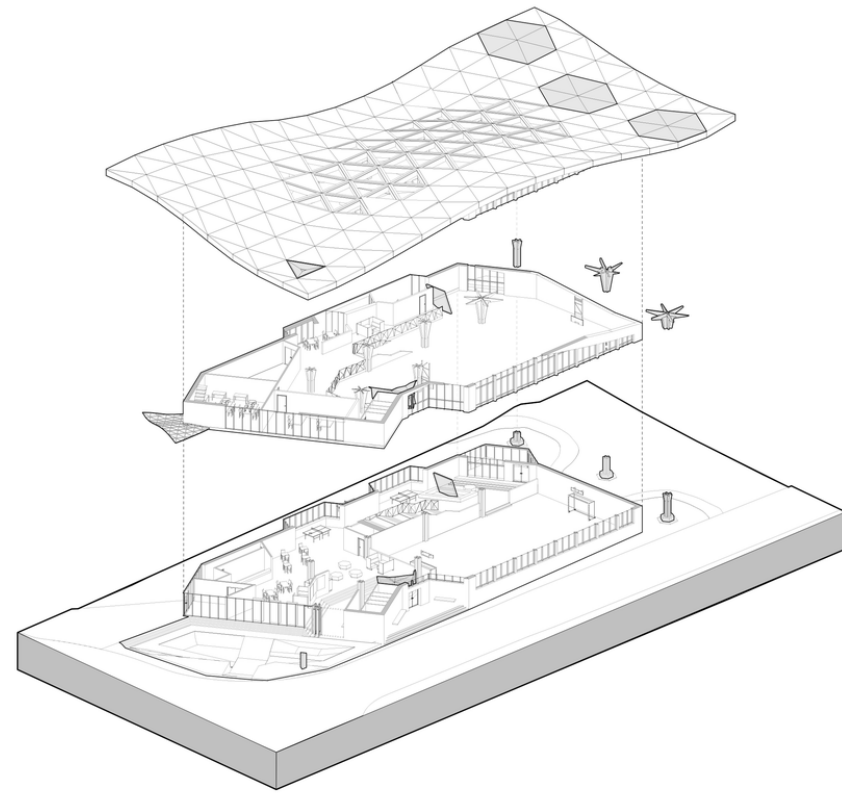


A shelter for the youth

It would have been easy to leave the roof as a shape that just looked right and let the structure figure itself out later, but I wanted to actually figure it out, get it to genuinely work.

A shelter for the youth carries a double meaning here, shelter from the weather but also shelter in a social sense, and the two aren't really separate. A lot of what exists for teenagers today only shelters them some of the time, cafes and malls that expect you to spend money to stay, youth centers that close before the evening really starts. Teens are usually left with whatever space is over, a parking lot, a bus stop, somewhere they've adapted rather than somewhere built for them. Getting the column and the grid to genuinely work was part of making that shelter real in both senses, not just a space that looks protected but one that actually is, whenever someone needs it.

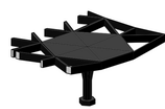




True youth scenarios

Activities in Rain Hub are based on personal experience and research from talking to youth leaders. It starts with a pickup football game, something I grew up with, everyone showing up in whatever clothes they've got, no kit, no organizing beforehand. Some people play the whole night, others drift off to a party halfway through and come back later, some don't play at all and just sit at the edge talking. It works because the ages are mixed and nobody's enforcing a schedule, the space just holds whatever version of the evening people want.

That's the kind of use Rain Hub is built for, not a single defined activity but a container loose enough for all of it to happen at once. The rain conversation area works the same way from a different angle, the sound of rain on metal loud enough to filter out everything else, so two people can talk privately without needing a wall between them and the rest of the hub. The rain display wall does something quieter, watching water move down it is meant to be calming, a spot to sit alone in the middle of everything else going on. None of these needed rain to be avoided, they needed it to be there.



Rain conversation area



Rain wall



Rain sculpture



Rain measuring station

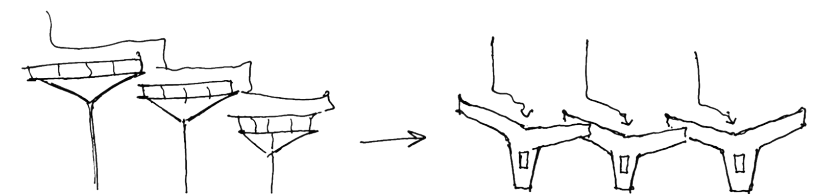
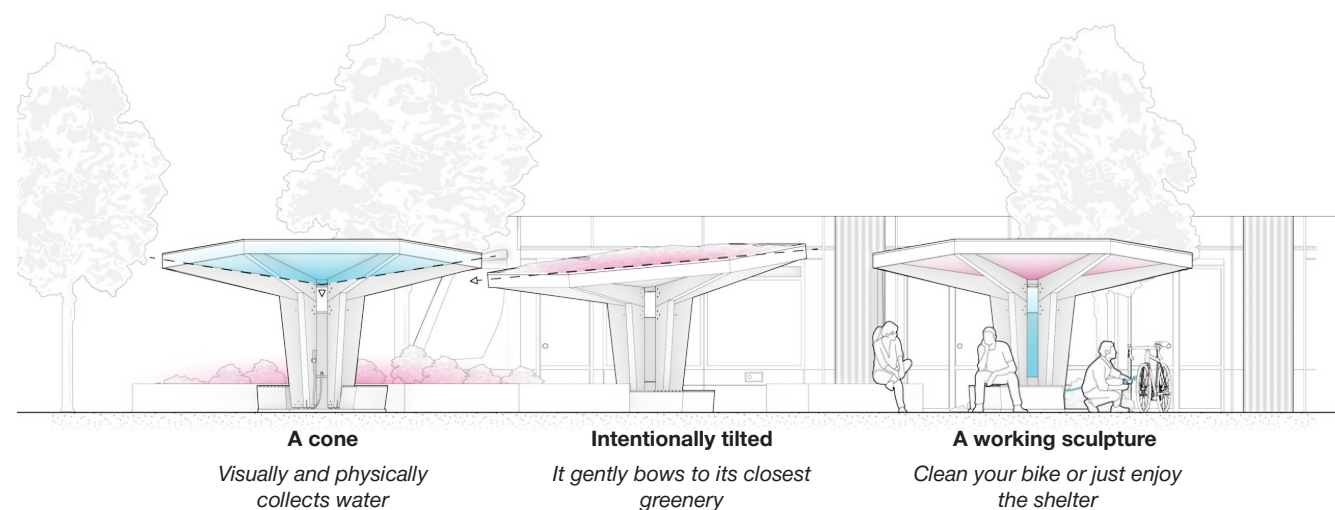




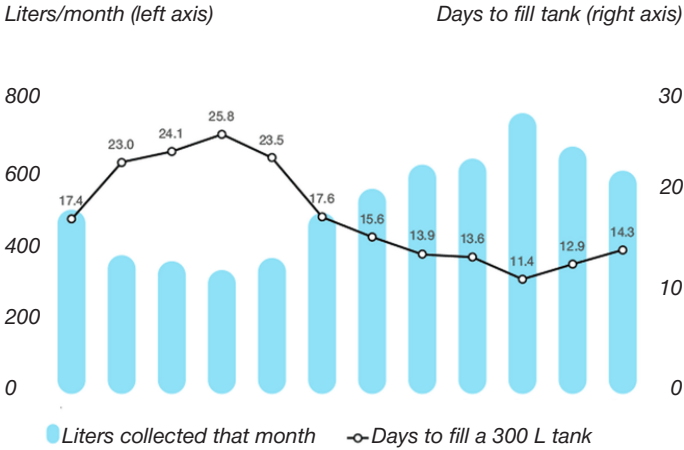
Campustratten

URBFORM 1:1 Competition winner 2026 - to be built
Duo work

Campustratten is meant to work as an evolution of, and a sibling to, the existing LTH fountain, something with a similar spirit but its own identity on campus. The roofs are designed to be visible from a distance, and the way their angles interact creates a slight imperfection, nothing perfectly aligned, which is exactly what makes them feel approachable rather than monumental. Placed the way they are on campus, they naturally become a meeting spot, somewhere to stop, sit in a circle, and actually have a conversation without needing a reason to be there. Rainwater gets gathered visibly by the roof and filtered down through the central column, so on a rainy day you can sit underneath and watch the water work its way through the filter, or just use what's been stored to wash your bike. Each roof reads as a cone, physically and visually pulling water toward its center, tilted slightly as if bowing toward the greenery closest to it. It ends up being three things at once, a landmark you can spot from across campus, a social node, and a working piece of infrastructure you can actually use.



An evolution and sibling to the existing LTH-fountain
A visually pleasing structure connected to the site and with a purpose

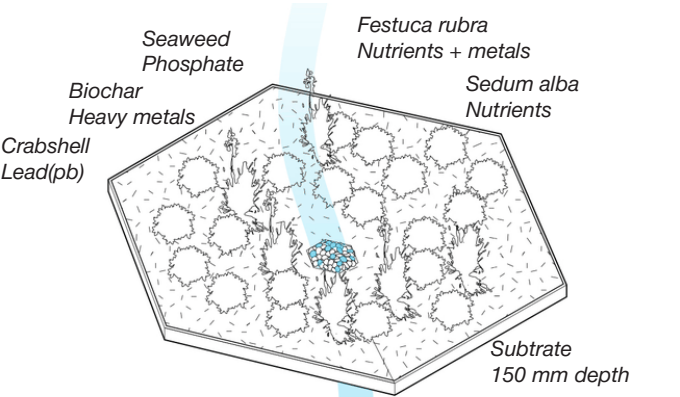


¹ Georgin, J. et al. (2026) 'Green Roof Substrates for Water Quality Improvement', *Molecules*, 31(11), 1862. doi:10.3390/molecules31111862.

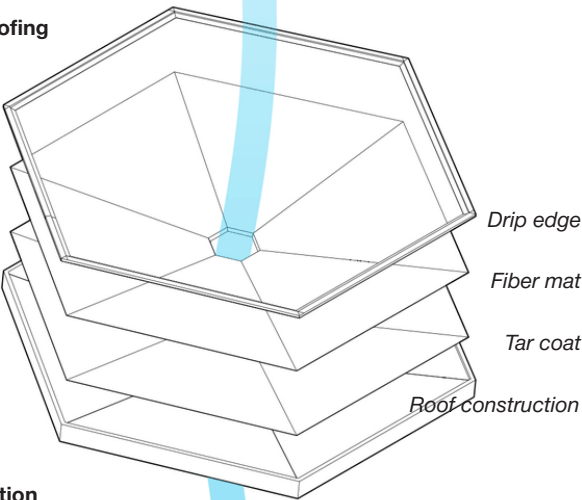
² Koivusalo, H. et al. (2023) Performance of Sand and Mixed Sand-Biochar Filters... *Water*, 15(8), 1631.

Green roof for improved runoff

Vegetation and substrate selected to absorb metals and enhance water quality.¹

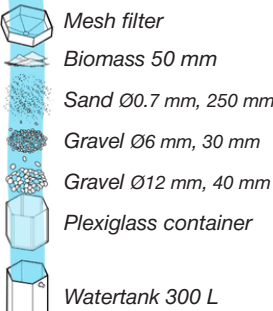


Waterproofing



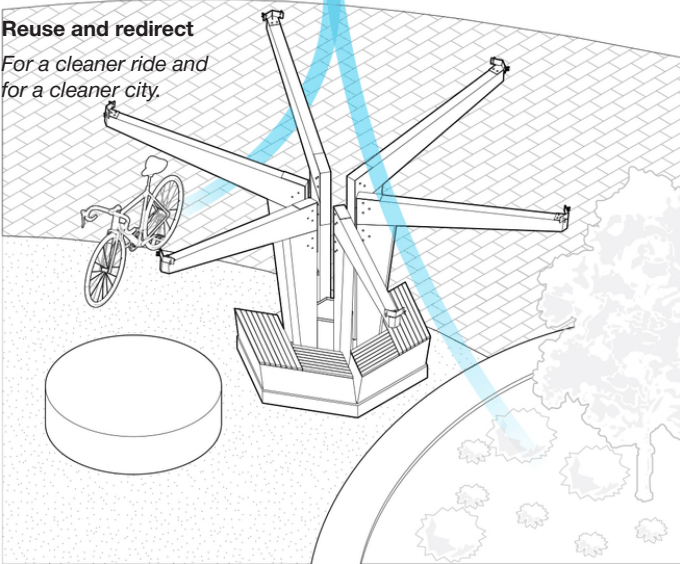
Fine filtration

Sand filter demonstrated to improve runoff quantity and quality.²



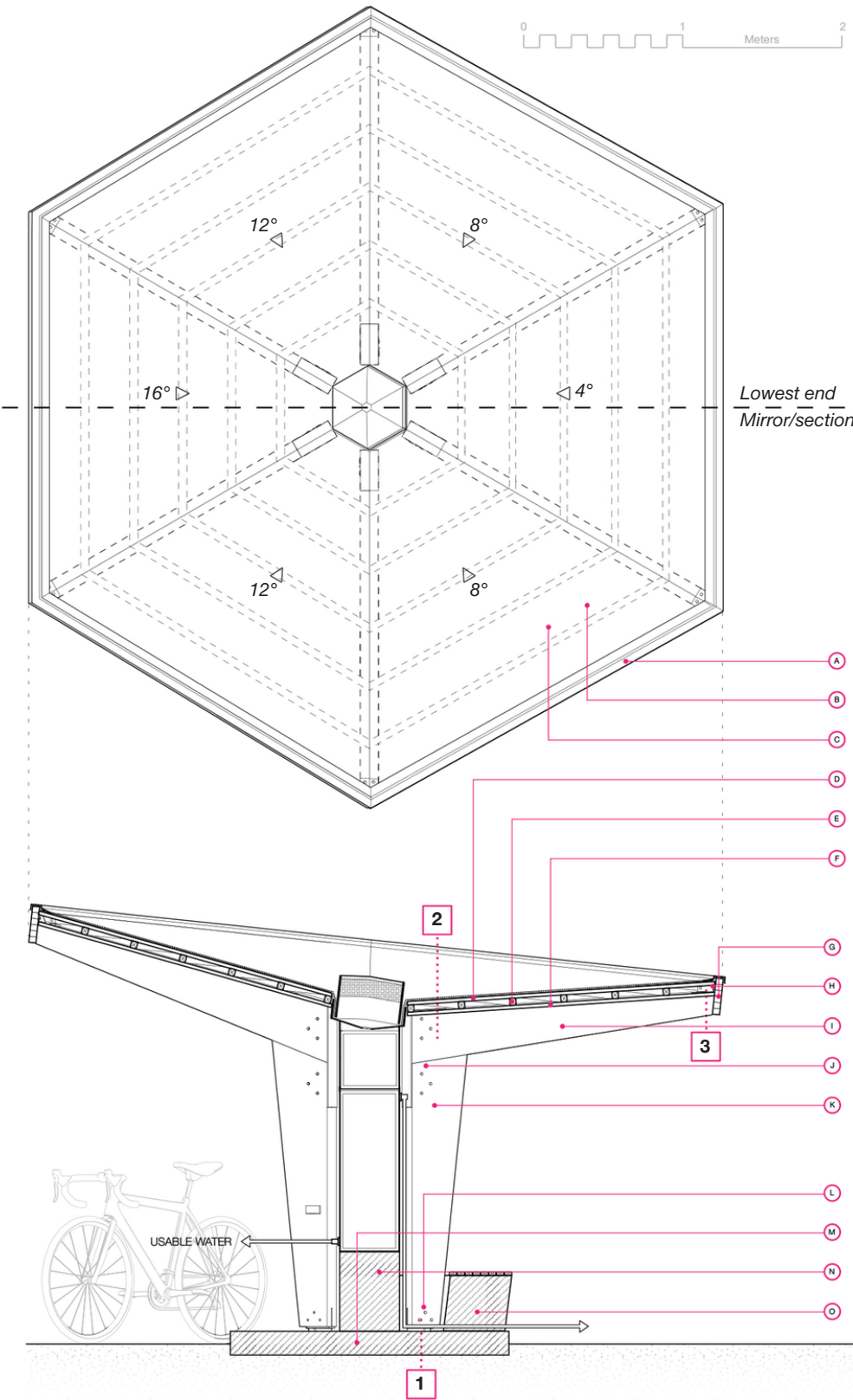
Reuse and redirect

For a cleaner ride and for a cleaner city.



Designed for buildability

The roof geometry is mirrored, which means the pillar and beam members always occur in pairs, so instead of fabricating six unique components, workers only have to make three components twice. That single decision cascades through the whole structural system, tapered glulam beams and pillars connected through welded steel plates, everything sized off rough load calculations covering dead load, snow, and the wet weight of the substrate on the roof. The foundation follows the same logic, precast concrete pads and blocks that also double as counterweight and seating, so nothing about the base is purely structural, it's doing at least two jobs at once. Getting this right meant thinking about a construction crew on site as much as about the geometry itself, because a structure this expressive only works if someone can actually put it together without every piece being a special case.



Roof structure

- A. 0.7 mm galvanized steel drip edge
- B. Fiber mat
- C. Tar coat
- D. 22mm plywood (upper sheathing)
- E. 45x45 mm pine battens
- F. 22 mm plywood (lower sheathing)

Structural system

- G. 45x225 mm structural fascia board
- H. 10 mm welded S355 steel plate
- I. Tapered GLT beam, 115x360-135 mm
- J. 10 mm welded S355 steel plate
- K. Tapered GLT pillar, 115x200-405 mm

Foundation

- L. 10 mm welded S355 steel plate, cast into precast concrete pad
- M. Precast concrete pad, 2000x150 mm
- O. Precast concrete seating block (additional weight)

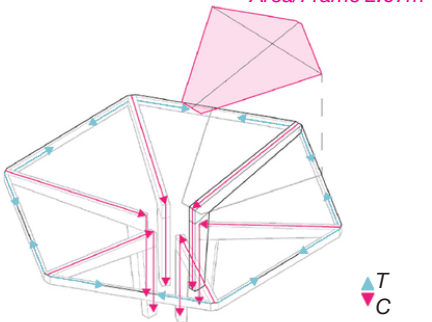
Load calculations

Rough calculations and simulations were made only to estimate the frame dimensions.

Roof load data

Substrate 150 mm
Wet load 150kg/m²
Area 16m²

Total Area 16m²
Area/Frame 2.67m²



Loads

2kN/m²

Dead Load (G_d)
Substrate wet (1.5kN/m²) + Self weight (0.5kN/m²)

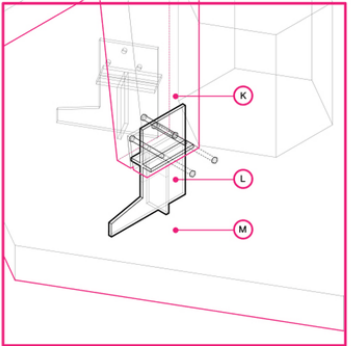
1kN/m²

Live load (Q_k)
Snow.

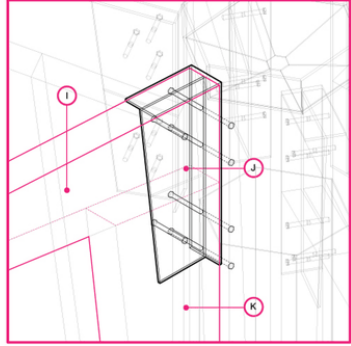
ULS Combinations

$q_d = 1.35 G_k + 1.50 Q_k = 4.2kN/m^2$
Governing

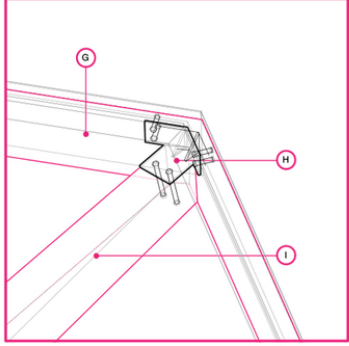
1. Base plate anchor

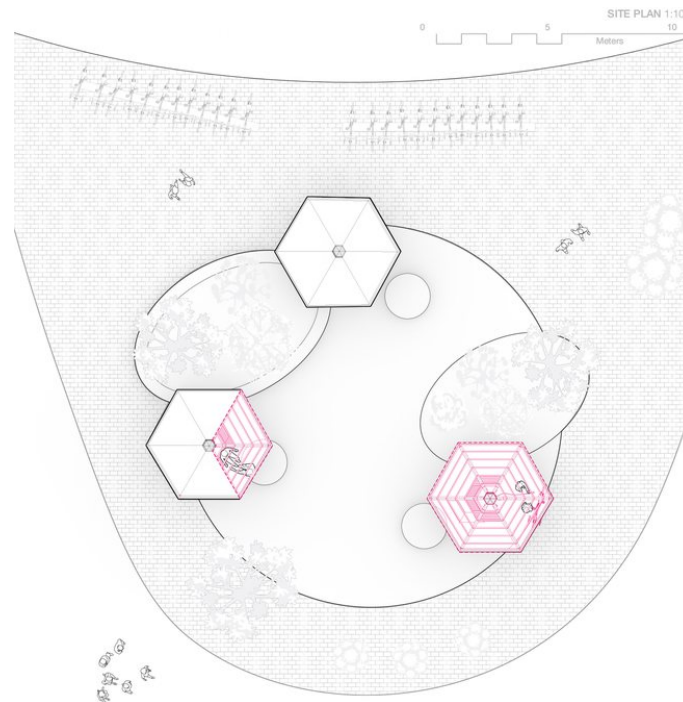


2. Column-to-beam bracket



3. Eave connection



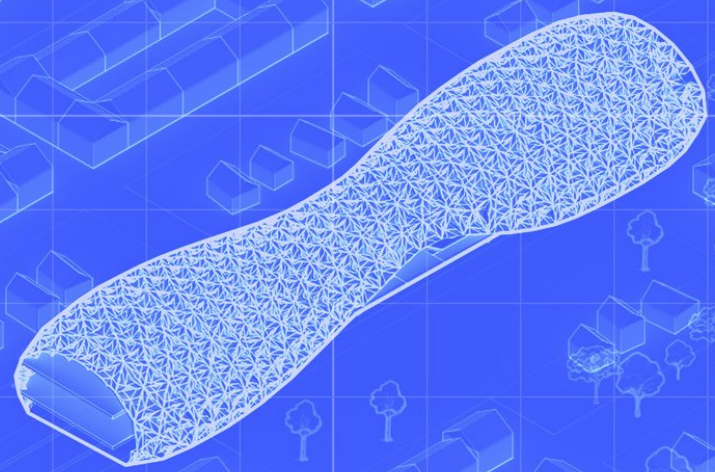


Social node by the Kårhus

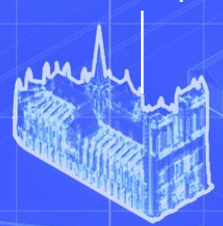
The placement wasn't incidental. Sitting outside the kårhus and next to the bus stop puts Campustratten right where movement across campus is already concentrated, people waiting, people passing through, people killing five minutes before they have somewhere to be. Rather than compete with that flow, the roofs give it somewhere to land, a spot to actually stop instead of just passing by or standing around waiting for a bus. It works because it doesn't ask for a reason to visit, you don't go to Campustratten specifically, you end up there on your way to or from somewhere else, and that's what makes it a genuine social node rather than a destination that has to justify itself.

The project ended up winning the competition, and we're currently in conversation with structural engineers to actually getting it built.





rhinocp



Artist studios



There is loads more

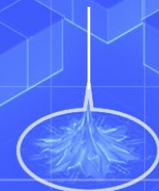
This is just a slice of it. I'd love to talk more about all these projects and everything else I have been working on. Reach out if you want to chat, or have a look at more on:

www.simonhildell.com

simon@hildell.com

[ig simonhildellworks](#)

GPT 6 Astra integration



Sport center



ByggCAD Complex Revit



Nakagin Unreal Engine

